

Divisiveness of Content. We say content with parameters (p', q') is *more divisive* than content with parameters (p, q) if $p \geq p'$ and $q \leq q'$. Divisive content has a message that is more tethered to the true state θ when it is truthful (and more likely to argue against θ if it is misinformation). In our case, we think of state θ as related to political ideology. Therefore, non-political content, such as wedding photos or cat videos, has little divisiveness relative to more political ones, such as “Obama Signs Executive Order Banning The Pledge of Allegiance in Schools Nationwide” (Fourney et al. (2017)). Note also that, from equation (2), prior beliefs about θ matter more for updating and the assessment of an article’s veracity when content is more divisive.

Polarization of Beliefs. We say H_2 is *more polarized* than H_1 if it satisfies the following single crossing property: $H_2^{-1}(\alpha) - H_1^{-1}(\alpha)$ is a nondecreasing function in α , crossing zero at $\alpha^* = 1/2$ with $H_1(1/2) = H_2(1/2) = 1/2$.

An increase in polarization results in a “stretching” of the belief distribution around the most moderate user (i.e., $b = 1/2$) while preserving an equal distribution of left-wing and right-wing agents (meaning that $H(1/2) = H(1/2) = 1/2$, which is applied in the island model to the average distribution of beliefs, $H = \frac{1}{N} \sum_{\ell=1}^N N_{\ell} H_{\ell}$). The available evidence indicates that the US public has become more polarized (see Pew Research Center (2014) and Abramowitz (2010)), and an important question of debate has been whether this polarization has fueled the spread of misinformation on social media.

Virality of Content. The next theorem studies how political divisiveness and polarization impact social media behavior and the spread of misinformation. To highlight our main insights, we focus on the same two regimes identified in Theorem 2 (low reliability with $r < \underline{r}$ and high reliability with $r > \bar{r}$) and in addition distinguish between high and low levels of homophily ($p_s/p_d > \bar{p}$ and $p_s/p_d < \underline{p}$):

Proposition 3. *There exists $r^* \in (0, 1)$ and $p^* \in (0, 1)$ such that:*

- (a) If $r < r^*$ and $p_s/p_d > p^*$, then an *increase* in divisiveness or an increase in polarization leads to greater content virality.
- (b) If $r > r^*$ and $p_s/p_d < p^*$, then a *decrease* in divisiveness or a decrease in polarization leads to greater content virality.

Proposition 3 builds on and is complementary to Theorem 2. When the content in question has high reliability ($r > \bar{r}$), more divisive content or greater polarization do not accentuate virality (and in fact may reduce content virality when homophily is limited, because in a well-connected, non-homophilic network, controversial articles will solicit a wide range of reactions, disciplining those tempted to share misinformation). In contrast, when the article in question has low reliability and there is significant homophily, there are again echo chamber-like effects. More divisive content generates more divergent behavior from individuals with different ideologies, and greater polarization means there are sharper differences in terms of these ideologies. As a result, echo chambers matter especially for divisive content and in the presence of polarization. The intuition of this result builds on Theorem 2: greater polarization can trigger more negative reactions, especially for divisive content and especially when there more polarized opinions. But when there is a high degree of homophily,